Table 1. DIRECTIONS OF MAGNETIZATION OF BASIC INTRUSIVE ROCKS
N, Number of samples; θ, radii of circles of confidence; $\hat{K}$, dispersion parameter; D, declination; I, inclination

	N	θ	$\hat{K}$	D	I
Mount Megantic	12	6·5	44	157	−52
Other Monteregian Hills	10	18°	7·9	340	+57
Other Monteregian Hills	39	9°	7·5	142	−62

Table 2. PALÆO POLE POSITIONS

	Latitude	Longitude
Monteregian Hills	65° N.	157° W.
Mount Megantic	68·5 N.	171·5 E.
Dakota sandstones*	74 N.	168 W.

* Pole position as computed by me from directions of magnetization and collecting sites given by Runcorn (1956).

Table 3

	Mineral in parent rock	Age in 10^6 yr.	Laboratory
Mount Royal	Pyroxene in Tinguaite	100	M.I.T.
Mount Johnson	Biotite in Essexite	110	M.I.T.
Mount Brome	Biotite in Nordmarkite	122	G.S.C.
Mount Megantic	Biotite in Gabbro	115	G.S.C.

Potassium/argon age determinations by Geological Survey of Canada (G.S.C.) and the Massachusetts Institute of Technology (M.I.T.).

basic rocks and the Monteregian Hills were intruded at the same time.

Pole positions derived from other North American rocks of known ages[4] are also plotted in Fig. 3. It appears from these data that a marked shift of the pole occurred between Cretaceous and Triassic times. It would seem, therefore, that pre- and post-Triassic rocks could be distinguished on the basis of their palæomagnetic directions. If this line of reasoning were legitimate, the '*M*' and '*m*' poles belong to the post-Triassic group and their proximity to the Cretaceous pole would suggest further a Cretaceous age for both the Mount Megantic and the Monteregian Hills rocks. This conjecture was recently supported by potassium/argon age determination data (Table 3) released by two independent laboratories on samples collected from three of the Monteregian Hills and from Mount Megantic. Thus, palæomagnetic and radiogenic data support independently the synchronous intrusions of these two rock units and place them in Cretaceous times.

I thank the Director of the Geological Survey of Canada for permission to publish this work.

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DIFFERENTIAL SECONDARY MAGNETIZATION IN SOME BRITISH CAMBRIAN ROCKS

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TWELVE orientated rock specimens were collected from three horizons in the Hartshill Quartzites and the Hyolithus Limestones from two localities near Nuneaton. Geologically[1] the quartzites include true quartzites and ferruginous sandstones coloured red or pink, while the limestones consist of organic material in a fine-grained matrix of calcite, iron minerals and other material, the whole strongly red stained. The rocks dip to the south-west at about 40 deg.

Magnetic results were obtained for eleven rocks and are plotted on the equal area stereonet (Fig. 1) in their field orientations; the twelfth was too weak to give significant results with the existing magnetometer. Combining the vectors for the four specimens from the Hyolithus Limestones by taking the mean of the direction cosines and making a statistical estimate of the dispersion after Fisher[2] gives:

$$D\ 017°\quad I\ 23°\quad \alpha\ 16°\quad \varkappa\ 33$$

The following abbreviations are used : D, declination measured from true north; I, inclination to the horizontal, positive downwards; α, semi-angle of the cone of confidence drawn at the 95 per cent probability-level; $\varkappa$, estimate of precision. In contrast, the mean direction obtained from the seven specimens from the quartzites is:

$$D\ 132°\quad I\ 64°\quad \alpha\ 25°\quad \varkappa\ 6$$

Correcting for the dip of the rocks moves both the mean directions towards the south-west by about 20 deg., so that when these directions are compared with those found in Cambrian rocks from Wales[3] the quartzites give a similar direction, but the mean vector for the limestones is about 90 deg. removed.

All the rocks were demagnetized in a.c. fields of up to 100 œrsteds using apparatus similar to that described by Creer[4]. The new vectors obtained on re-

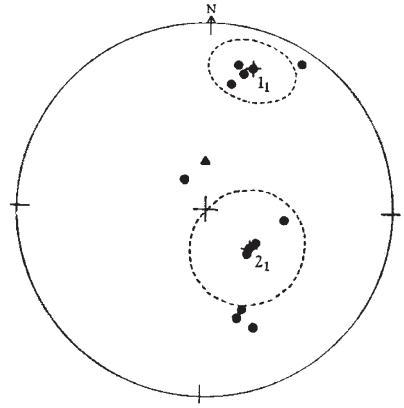


Fig. 1. Equal area stereonet showing the positions of the magnetic vectors for eleven rocks from the British Cambrian together with the group mean vectors. All vectors are in field orientation. ●, Vector plotting in lower hemisphere; 1_1, mean vector for limestones; 1_2, mean vector for limestones after a.c. washing; 2_1, mean vector for quartzites; 2_2, mean vector for quartzites after a.c. washing; 2_3, mean vector for quartzites after a.c. washing and correction for dip; •, vector for Trias rocks; ■, vector for Cambrian rocks; ▲, direction of present dipole field. The broken lines show the circles of confidence drawn at the 95 per cent level

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measuring (Fig. 2) give mean directions for the quartzites:

$$D\ 161^\circ \quad I\ 55^\circ \quad \alpha\ 15^\circ \quad \varkappa\ 14$$

and for the limestones:

$$D\ 015^\circ \quad I\ 44^\circ \quad \alpha\ 9^\circ \quad \varkappa\ 115$$

where the rocks are in their field orientations. Both groups show a reduction in the amount of scatter and both show a displacement of the direction of the mean towards the south-west which cannot be attributed to the removal of a component of secondary magnetization parallel with the present geomagnetic field.

The displacement of both mean vectors in a similar direction would also rule out the possibility that the limestones were originally magnetized in a reverse direction to the quartzites and have both taken on a component of secondary magnetization to give their present positions. The presence of a reversal is also unlikely since the limestones occur stratigraphically within the quartzite series, samples from above and below the limestone horizon give similar directions of magnetization and there is no evidence of any marked break in deposition.

Comparing the change in the intensities of magnetization produced by the a.c. washing there is a marked difference between the limestones, which lose about 85 per cent, and the quartzites, which show little change in intensity. This suggests that the two rock types have taken on very different components of secondary magnetization.

The mean vectors for the two groups in field orientation are given in Fig. 3 both before and after a.c. washing together with mean vectors for rocks of Trias age from Europe[5,6] and for the Caerbwdy Sandstones of Cambrian age from Wales[3]. The mean direction for the Hyolithus Limestones before washing lies within the group of Trias results and accords well with those from rocks of Bunter age; the effect of the washing is to shift the mean direction towards the Cambrian direction. For the Hartshill Quartzites the mean before washing lies close to the Cambrian direction, and the washing moves the vector still closer so that if this mean direction is corrected for the observed dip of the rocks the resultant vector:

$$D\ 193^\circ \quad I\ 34^\circ \quad \alpha\ 16^\circ \quad \varkappa\ 12$$

is in good accord with the Cambrian direction.

Combining the magnetic and geological evidence these observations may be explained as follows: The Hartshill Quartzites and the Hyolithus Limestones were weakly magnetized parallel to the geomagnetic field in Cambrian times. Uplift and erosion having removed any overlying rocks, these Cambrian rocks lay at or near the surface in the Trias. During the deposition of the thick series of red beds of Trias age which are now exposed to the north-west of the Cambrian rocks, these were penetrated by iron-bearing ground waters which, in depositing new iron minerals and oxidizing existing minerals to new forms, imposed a chemical remanent magnetization[7,8]. The degree to which the Cambrian rocks were affected by this process depended on their permeability; this was slight in the quartzites, but greater in the more permeable limestones. That this new magnetization of the limestones could not be removed completely by washing in a.c. fields up to 100 œrsteds is in agreement with Creer's work[4] on unstable Keuper Marls, which in some cases still retained components of secondary magnetization after washing in fields of up to 850 œrsteds.

If it is correct to assume that the quartzites after a.c. washing give a true record of the direction of the Cambrian geomagnetic field the position of the magnetic pole at that time can be calculated as:

$$\text{Lat. } 18^\circ \text{ N.} \quad \text{Long. } 165^\circ \text{ E.} \quad d\psi\ 10^\circ \quad d\chi\ 18^\circ$$

where $d\psi$ and $d\chi$ are the lengths of the semi-axes of the elipses of confidence drawn at the 95 per cent probability-level. This pole position is in good agreement with those determined from the British Cambrian and from rocks of similar age from Norway[9].

This work was carried out while I was a student at the Sedgwick Museum, Cambridge, and I take this opportunity to thank W. B. Harland and Dr. R. Hey of that department for their help and encouragement.

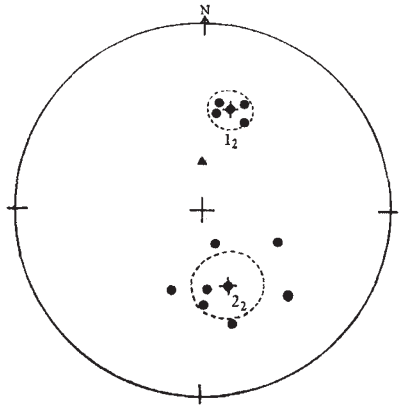


Fig. 2. Equal area stereonet showing the positions of the magnetic vectors after a.c. washing, together with the group mean vectors. All vectors in field orientation. Symbols as in Fig. 1

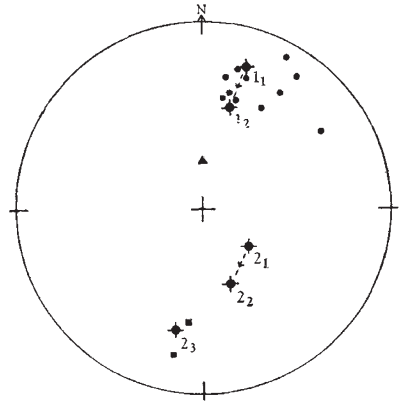


Fig. 3. Equal area stereonet showing the relation of the mean positions for these Cambrian rocks to directions determined by other workers from Trias and Cambrian rocks. Symbols as in Fig. 1

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